HEREDITARY DISORDERS OF GENOME INSTABILITY AND DNA REPAIR

AT A GLANCE

- Genome instability characterizes a large group of inherited disorders with prominent cutaneous abnormalities; many have an increased cancer risk.
- Genome instability is caused by impaired repair and/or maintenance of DNA.
- **Xeroderma pigmentosum (XP)** is a prototype disorder with impaired repair of environmentally induced DNA damage and a greatly increased frequency of sunlight-induced cancers.
- Although these disorders are rare, heterozygote carriers of affected genes—potentially comprising several percent of the general population—may also have an increased risk of neoplasia.

Overview

DNA serves as the carrier of genetic information and is constantly threatened by damaging agents such as oxidative stress, ultraviolet (UV) radiation, X-radiation, and chemical mutagens. While cells possess robust mechanisms to repair most DNA damage, failure to maintain genomic integrity can lead to abnormal cell function, cell death, or malignant transformation. If a damaged cell divides, its progeny may accumulate additional mutations, and progressive genomic instability can ultimately result in cancer.

The accumulation of multiple mutations in key genes within a single cell is required for malignant transformation. A **mutator phenotype**, characterized by increased mutation rates due to genome instability, is often considered a prerequisite for carcinogenesis. Without such instability, the probability of acquiring all necessary oncogenic mutations in one cell would be exceedingly low.

Genes implicated in hereditary genome instability syndromes can also contribute to sporadic cancers when somatically mutated, highlighting their critical role in suppressing tumorigenesis from an early stage.

Although these inherited disorders are rare (with incidences on the order of 10^{-5} to 10^{-6}), heterozygous carriers of pathogenic alleles may represent several percent of the general population. Most of these conditions follow autosomal recessive inheritance, so carriers are typically asymptomatic. However, epidemiological evidence suggests that carriers may still face a modestly increased risk of cancer.

Patterns of Genome Instability

Genome instability manifests differently across syndromes:

- In **Bloom syndrome (BS)**, **ataxia-telangiectasia (A-T)**, and **Fanconi anemia (FA)**, spontaneous chromosomal breakage is observed in primary blood or skin cells, indicating intrinsic defects in DNA maintenance.
- In **xeroderma pigmentosum (XP)**, genome instability is primarily evident only after exposure to DNA-damaging agents such as UV radiation or chemical carcinogens like benzo[a]pyrene (found in cigarette smoke). Cells from XP patients show heightened sensitivity to UV-induced DNA damage due to defective nucleotide excision repair (NER).

The clinical severity in XP—particularly the high incidence of skin cancers and corneal scarring leading to blindness—results from the interaction between genetic susceptibility and environmental exposure. Notably, XP patients who rigorously avoid UV radiation can dramatically reduce or even prevent the development of skin cancer and ocular complications.

DNA Repair Pathways and Disease Mechanisms

Several DNA repair pathways protect against specific types of damage:

- **Nucleotide excision repair (NER)**: Repairs bulky DNA adducts caused by UV light (e.g., cyclobutane pyrimidine dimers). Defective in XP.
- **Base excision repair (BER)**: Corrects small base lesions caused by oxidation or alkylation.
- **Double-strand break repair (via homologous recombination and non-homologous end joining)**: Impaired in A-T and FA.
- **Interstrand crosslink repair**: Defective in FA.
- **DNA damage response signaling**: Disrupted in A-T (ATM gene) and BS (BLM helicase).

Defects in these pathways lead to characteristic cellular phenotypes such as hypersensitivity to DNA-damaging agents, increased mutagenesis, and chromosomal instability.

TABLE 140-1 Hereditary Disorders of Genome Instability and DNA Repair

DISORDER	CUTANEOUS	NEOPLASIA	OTHER	INHERITANCE	CELLULAR ABNORMALITIES	AFFECTED GENE(S)	IMPAIRED FUNCTION
XP	Photosensitivity, poikiloderma (hyper- and hypopigmentation, atrophy, telangiectasia), skin cancers	BCC, SCC, melanoma, central nervous system tumors	Sensorineural deafness, progressive neurologic degeneration, primary loss of neurons (in some)	Autosomal recessive	Increased UV-induced cell killing and mutagenesis	XPA, XPB, XPC, XPD, XPE, XPF, XPG	Nucleotide excision repair (NER)
Cockayne syndrome (CS)	Photosensitivity, cutaneous atrophy, lack of skin cancer despite severe photosensitivity	Rare	Severe growth failure, neurological dysfunction, cataracts, premature aging	Autosomal recessive	UV hypersensitivity, defective transcription-coupled NER	CSA, CSB, others	Transcription-

coupled repair, mitochondrial function | | **Trichothiodystrophy (TTD)** | Ichthyosis, brittle hair and nails, photosensitivity | No increased risk | Intellectual disability, short stature, infertility | Autosomal recessive | UV sensitivity, reduced DNA repair capacity | XPB, XPD, TTDA | NER and basal transcription | | **Bloom syndrome (BS)** | Photosensitivity, facial erythema (butterfly distribution), telangiectasia | Early-onset cancers (e.g., leukemia, lymphoma, GI cancers) | Growth deficiency, insulin resistance, immunodeficiency | Autosomal recessive | Increased sister chromatid exchanges, chromosomal breakage | BLM (RECQL3) | DNA helicase function, replication fork stability | | **Ataxia-telangiectasia (A-T)** | Oculocutaneous telangiectasia, premature graying, skin atrophy | Leukemia, lymphoma, solid tumors | Cerebellar ataxia, immunodeficiency, pulmonary disease | Autosomal recessive | Radiosensitivity, chromosomal instability, defective cell cycle checkpoint | ATM | DNA damage response signaling | | **Fanconi anemia (FA)** | Hyperpigmentation, café-au-lait spots, hypopigmentation, skin atrophy | AML, head and neck SCC, gynecologic cancers | Congenital malformations, bone marrow failure | Autosomal recessive or X-linked | Increased chromosomal breakage with crosslinking agents | FANCA, FANCB, FANCC, ..., FANCP/ SLX4 | Interstrand crosslink repair, homologous recombination |

> **Note:** BCC = basal cell carcinoma; SCC = squamous cell carcinoma; AML = acute myeloid leukemia.

Clinical Relevance and Dermatologic Involvement

All of these disorders present with significant cutaneous manifestations, often prompting initial evaluation by dermatologists. Skin findings may include photosensitivity, pigmentary changes, atrophy, telangiectasia, and premature aging. Importantly, the skin serves as a visible indicator of systemic genomic instability.

Dermatologists play a key role in early diagnosis, cancer surveillance, patient education (especially regarding sun protection), and multidisciplinary management. Given the elevated cancer risk, regular screening and preventive strategies are essential.

Moreover, understanding the molecular basis of these rare syndromes provides critical insights into the broader mechanisms of DNA repair, aging, and cancer development in the general population.